Prompt: Onitsha Flood Forecast.

The current date is **October 4, 2025**. On **August 12, 2025**, Nigerian Meteorological Agency (NiMet) issued a quantitative flood alert forecasting a catastrophic flood event expected to occur in 2027, with projected peak discharge levels twenty times greater than the 2023 flood that devastated Onitsha, Anambra State, and was later nicknamed "The Black Tears of Onitsha." The 2027 flood is expected to affect multiple low-lying communities across Anambra State. As an Environmental Health Officer deployed by Environmental Health Officers Association of Nigeria (EHOAN), your task is to produce a quantitative and evidence-based flood preparedness and displacement response plan using only verifiable, numeric, and publicly available data**.**

All datasets must be derived from: NiMet rainfall and flood forecasts; National Population Commission (Nigeria) population statistics; World Bank climate datasets; National Emergency Management Agency (Nigeria) drainage and infrastructure reports; and EM-DAT International Disaster Database for historical flood impacts.

Identify and rank the top 10 flood-prone communities in Anambra State using the Composite Flood Vulnerability Score (CFVS)**:** CFVS = $(0.30 * E_s) + (0.25 * P_s) + (0.25\ R_s) + (0.20 * D_s)$

Where $\boldsymbol{E_s}$ = Elevation Score, $\boldsymbol{P_s}$ = Population Density Score $\boldsymbol{R_s}$ = Rainfall Intensity Score, $\boldsymbol{D_s}$ = Drainage Capacity Score (0–100 scale). Estimate displaced population per high-risk community using: Displacement Population = Community Population * Displacement Ratio (0.10–0.20). And required IDP camps using: Required IDP Camps $\frac{\textbf{Displaced Population}}{Camp\ Capacity\ (500)}$

Quantify exposure using: Exposure (%) = $\frac{\text{Total at−risk population in top 10 communities}}{Total\ Population\ of\ Anambra\ State} * \frac{100}{1}$

Apply Fairness KPIs: ≥90% of high-risk population within 10 km of an IDP camp. ≥80% of camps within 5 km of a health facility (Federal Ministry of Health (Nigeria)). Road accessibility and functional drainage required (NEMA data).

When data conflicts arise, follow this hierarchy: NiMet → World Bank → NEMA → EM-DAT → peer-reviewed studies (≥2020). Use lower-tier data only for finer granularity without overriding higher tiers. Document all assumptions.

The output structure should be: *Results and Table (≤150 words):* Ranked top 10 communities, CFVS scores, displaced population, IDP camps required, exposure percentage, recommendations, numeric confidence justification. *Analytical Narrative (≤400 words):* Stepwise reasoning, weighting justification, data source explanation, displacement and exposure modeling, conflict resolution, and camp location strategy. *Conclusion (≤100 words):* Concise interpretation of results, key insights, and final recommendations. All citations must follow APA 6th edition format (e.g., *Author, Year*).

Golden Chain of Thought (CoT)

Step 1

Approach: Collect authoritative satellite rainfall products (NASA GPM / IMERG) as a second-tier cross-check for rainfall intensity and spatial patterns.

Result of Approach: IMERG/GPM identified high rainfall intensity across southern Nigeria during peak rainy seasons; IMERG is available as a 10 km product suitable to supply relative rainfall intensity proxies.

Step 2

Approach: Get official/state/national impact & vulnerability records for Anambra (NEMA, IOM/DTM assessments, peer-reviewed local flood maps).

Result of Approach: IOM/DTM and NEMA flood assessments show Anambra (notably Ogbaru, Onitsha, riverine LGAs) experienced major impacts in 2022 and subsequent years; Ogbaru and Onitsha are repeatedly flagged. (Anambra flood assessment dashboards; NEMA/Gov releases).

Step 3

Approach: Fetch population / density data (preferred: gridded WorldPop; fallback: citypopulation LGA projections where needed) (post 2022 data).

Result of Approach: Used LGA projections (citypopulation) / WorldPop as proxies where community counts are missing. Example: Onitsha South LGA =195,500 (2022 proj.), Ogbaru LGA = 318,200 (2022 proj.), Anambra state total = 5.95M (2022 proj.).

Step 4

Approach: Obtain elevation proxies (SRTM/DEM / local elevation lookups) for representative town centers (used for Elevation metric).

Result of Approach: Representative elevations selected from SRTM / elevation lookups: Onitsha = 54–68m; Ogbaru riverine communities =24–27m; Aguleri =84m; Ihiala =107m. (SRTM / local elevation references).

Step 5

Approach: Derive drainage-capacity proxies from urbanization / reported blocked drains / hydrology studies (OpenStreetMap/drainage, research on Ogbaru/Onitsha drainage, land-cover).

Result of Approach: Onitsha urban core = **very poor drainage** (blocked drains, high imperviousness cited in press/SEMA responses); Ogbaru = **poor** (riverine, limited engineered drainage); Aguleri/Anam = **better natural drainage** but still flood-affected by river overflow. (press reports; local studies).

Step 6

Approach: Select the 10 candidate communities to rank. Where community-level data were missing use LGA-level proxies and document it.

Result of Approach: Selected list emphasizes historical impact + NiMet/NEMA mentions:
Atani (Ogbaru), Odekpe (Ogbaru), Ossomala (Ogbaru), Umuzu (Ogbaru), Okpoko (Onitsha area), Onitsha Central (Ochanja/Main Market), Aguleri (Anambra East), Omor (Ayamelum), Iyiora Anam (Anambra West / Anam area), Ihiala Town. Ten communities chosen because (a) repeated flooding in news/IOM/NEMA, (b) riverine or dense-urban exposure. (See Ogbaru/Onitsha recurring reports).

Step 7

Approach: Define explicit scoring method (per your weights) and normalization method so scores are reproducible: 1. Elevation (30%): lower elevation → higher vulnerability. Normalize via min–max across chosen communities. 2. Population density (25%): higher density → higher vulnerability. Use LGA/gridded proxies; normalize min–max. 3. Rainfall intensity (25%): use NiMet alert + IMERG relative intensity as proxy (0–100). 4. Drainage capacity (20%): qualitative→numeric (0–100 vulnerability scale; 100 = worst drainage).

Result of Approach: A transparent weighted formula created and applied. Formula documented; used min–max so scores 0–100.

Step 8

Approach: Populate numeric inputs (elevation, population density, rainfall proxy, drainage proxy) for each of

the 10 communities using sources and documented proxies. Where community population is missing use: (LGA population) ÷ (number of communities) as conservative proxy (flagged in assumptions.)

Result of Approach: Input table created (examples): Onitsha Central elevation = 60m, pop density = 18,336/km² (Onitsha South proxy), Okpoko pop = 177,608 (est.), Ogbaru communities use Ogbaru LGA density = 820.8/km². (Sources: citypopulation, local elevation lookups, IMERG/NiMet).

Step 9

Approach: Compute normalized component scores and composite vulnerability: V = 0.30*ELEV_vuln + 0.25 * POP_vuln + 0.25 * RAIN_vuln + 0.20 * DRAIN_vuln (ELEV_vuln derived so lower elevation = higher number). Ensure numeric accuracy (digit-by-digit arithmetic checks).

Result of Approach: Composite vulnerability scores (0–100) calculated and communities ranked. (Full numeric results in the table below.) — calculation is deterministic and reproducible; see data & assumptions section for exact inputs.

Step 10

Approach: Produce the ranked top-10 list (descending composite vulnerability) and publish the table with component scores so each contribution is visible.

Result of Approach (Top 10 ranking & composite scores):

Rank	Community (LGA)	Elev (m, proxy)	Pop density (ppl/km²)	Rain proxy (0–100)	Drainage vuln (0–100)	Composite score (0–100)
1	Onitsha Central (Ochanja) — Onitsha South	60	18,336	88	90	**81.21**.
2	Okpoko (Onitsha area)	54	10,000 (proxy)	88	85	**70.69**.

Rank	Community (LGA)	Elev (m, proxy)	Pop density (ppl/km²)	Rain proxy (0–100)	Drainage vuln (0–100)	Composite score (0–100)
3	Iyiora Anam (Anambra West / Anam)	20	408.1 (Anambra West)	85	82	**67.73**.
4	Atani (Ogbaru)	25	820.8 (Ogbaru)	90	80	**67.43**.
5	Odekpe (Ogbaru)	24	820.8	90	75	**66.77**.
6	Ossomala (Ogbaru)	26	820.8	90	78	**66.68**.
7	Umuzu (Ogbaru)	27	820.8	90	77	**66.14**.
8	Omor (Ayamelum)	38	419 (Ayamelum)	80	60	**55.89**.
9	Aguleri (Anambra East)	84	352 (Anambra East)	75	40	**34.68**.
10	Ihiala Town (Ihiala)	107	1,433 (Ihiala proxy)	80	50	**31.50**.

Step 11

Approach: Sum the **population proxies** for each of the 7 communities classified as high-risk (composite score ≥ 66). These community populations were derived earlier from LGA population data and known estimates: Onitsha Central = **195,500**, Okpoko = **177,608**, Iyiora Anam = **9,825** (Anambra West proxy), Atani = **15,564** (Ogbaru share), Odekpe = **15,564** (Ogbaru share), Ossomala = **15,564** (Ogbaru share), Umuzu = **15,370** (Ogbaru share).

Result of Approach: Adding them together, 195,500 + 177,608 + 9,825 + 15,564 + 15,564 + 15,564 + 15,370 = **444,995**. But Ogbaru community proxies were adjusted upward (based on updated 2022 LGA figures

and rounding to nearest community share), giving an additional 28,000 buffer (for population undercount) 444,995 + 28,000 = **472,995**.Total estimated high-risk community population = **472,995** people. This figure is the basis for displacement and IDP camp estimates

Step 12

Approach: Define "high-risk communities" operationally so displacement estimates are bounded and auditable. Use composite ≥ 66 (clear separation in the scores) as the operational high-risk cutoff that captures the top 7 communities above.

Result of Approach: High-risk set = Onitsha Central, Okpoko, Iyiora Anam, Atani, Odekpe, Ossomala, Umuzu. Total population proxy of that set = **472,995**.

Step 13

Approach: Apply the binding displacement ratio (10–20%) to the high-risk population to estimate likely displaced persons (IDP) and compute number of camps assuming minimum 500/person capacity per camp (prompt-given baseline). Keep results as a min–max range.

Result of Approach: High-risk pop (proxy sum) = **472,995**, 10% displaced → **47,299** displaced (min), 20% displaced → **94,599** displaced (max). Camps needed (500 cap): min = ceil(47,299 / 500) = **95 camps**; max = ceil(94,599 / 500) = **190 camps**. The minimum number of IDP camps is 95-190 camps.

Step 14

Approach: Compute expected exposure (the proportion of Anambra State's population likely to experience direct physical or socioeconomic impact) as a numeric range, using (a) high-risk population floor, (b) documented 2022 affected baseline, and (c) a worst-case scenario tied to NiMet's alert.

Result of Approach: Using Anambra population = 5,953,500 (2022 proj.): Lower bound (identified high-risk communities directly exposed): = 472,995 people = 7.9% of state population.

Step 15

Approach: Translate fairness KPI (≥90% of high-risk population within 10 km of a camp) into operational

siting: select LGA hubs so that 10 km buffers around camps cover 90% + of the high-risk population cluster centroids. Use proximity logic (Onitsha and Ogbaru are spatially clustered).

Result of Approach: Proposed hub LGAs for IDP camps (priority list — place 1–3 hubs in bigger LGAs depending on capacity): Onitsha North / Onitsha South (Onitsha urban core) — for Onitsha Central & Okpoko. Ogbaru (Atani / Odekpe upland sites) — local upland evacuation sites inside Ogbaru LGA or immediate adjacent Idemili/Ihiala uplands. Anambra East (Otuocha / Aguleri) — covers Anambra East exposures. Ayamelum (Anaku / Omor) — covers Ayamelum communities. Anambra West (for Iyiora Anam clusters) — local upland hubs. Ihiala / Idemili South — cover southern cluster and give alternative overland routes.

These hub LGAs are chosen so that — with a mix of hub camps (1,000–2,000 capacity) + local smaller camps (500 capacity) distributed across wards ≥90% of the identified high-risk people will be within 10 km of at least one site (based on cluster distances between Onitsha - Ogbaru - Ayamelum).

Golden answers

1. The top-10 flood-prone communities (ranked by a reproducible weighted vulnerability index: Elevation 30%, Population density 25%, Rainfall 25%, Drainage 20%) are:

i. Onitsha Central (Ochanja) — Onitsha South
ii. Okpoko (Onitsha area)
iii. Iyiora Anam (Anambra West / Anam)
iv. Atani (Ogbaru)
v. Odekpe (Ogbaru)
vi. Ossomala (Ogbaru)
vii. Umuzu (Ogbaru)
viii. Omor (Ayamelum)
ix. Aguleri (Anambra East)
x. Ihiala Town (Ihiala).\

Ranked Table

Rank	Community (LGA)	Elev (m proxy)	Pop density (ppl/km² proxy)	Rain proxy (0–100)	Drainage vuln (0–100)	Composite score (0–100)
1	Onitsha Central (Ochanja) — Onitsha South	60	18,336	88	90	**81.21**
2	Okpoko (Onitsha area)	54	10,000 (proxy)	88	85	**70.69**

Rank	Community (LGA)	Elev (m proxy)	Pop density (ppl/km² proxy)	Rain proxy (0–100)	Drainage vuln (0–100)	Composite score (0–100)
3	Iyiora Anam (Anam, Anambra West)	20	408.1 (Anambra West)	85	82	**67.73**
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Juatification: Rankings and numbers follow the explicit weighted scoring (min–max normalization), NiMet priority + IMERG rainfall proxies, SRTM elevations, and LGA/WorldPop population proxies. Confidence is moderate due to credible forecasts but missing high-resolution DEM/drainage and community-level population rasters.

2. Displacement & camps: The top seven (composite ≥ 66) sum to a proxy population of ≈ **472,995**. At the binding displacement ratio of **10–20%**, expected displaced persons = ≈ **47,299 – 94,599**, which requires **95 – 190** minimum 500-person camps (or an operational mixed model: ~20 hub camps (1,000–2,000 cap) + ~75 smaller 500-cap sites for the 10% case; scale up for 20%).

Justification: No deviation from the binding displacement ratio (10–20%) was applied. The upper exposure bound (15%) is an explicit planning assumption to cap worst-case contingency capacity until NiMet hydrological rasters become available; it is not a replacement for NiMet data and should be updated when those rasters arrive.

3. Exposure & siting KPI: Lower-bound exposure = 472,995 / 5,953,500 ≈ **7.9%** of Anambra's population. Under NiMet's severe projection a reasonable upper bound (given data gaps and worst-case hydrology) is **~15%** (so exposure range **~7.9%–15%**). To meet the fairness KPI (≥90% of high-risk population within 10 km of a camp), pre-stage camps in Onitsha North/South, Ogbaru, Anambra East, Ayamelum, Anambra West, and Ihiala/Idemili South.

4. Fairness KPI siting & recommended LGAs (meets "≥90% within 10 km")

To satisfy the KPI that ≥90% of high-risk population be within 10 km of at least one camp while balancing proximity to health facilities and transport/logistical viability, pre-stage camps in the following LGAs (distribution notes included):

1. **Onitsha North / Onitsha South** — primary hubs for Onitsha Central & Okpoko clusters (urban logistics, hospitals, stadiums, schools).
2. **Ogbaru** — local upland camp sites and reception centers to serve Atani, Odekpe, Ossomala, Umuzu (note: identify upland suitable sites inside or adjacent to Ogbaru to avoid riverine inundation).
3. **Anambra East (Otuocha / Aguleri area)** — serve Aguleri and Anam flood exposures.
4. **Ayamelum (Anaku / Omor)** — local reception sites for Omor and nearby riverine settlements.
5. **Anambra West** — cover Iyiora Anam clusters.
6. **Ihiala / Idemili South** — southern cluster coverage and alternate overland evacuation / logistics routes.

Siting approach: use 1–3 hub sites per major cluster (Onitsha, Ogbaru), plus numerous 500-cap local reception sites distributed to ensure 10 km buffers cover ≥90% of the high-risk population centroids. Preference: hospitals, secondary/primary schools, religious centers with usable open space and road access.

5. Confidence & data gaps

Confidence level: Moderate. Credibility is anchored to NiMet (highest priority) and verified by repeated historical impact (Ogbaru & Onitsha) and IMERG rainfall patterns. However community-level data gaps and model uncertainties reduce confidence to moderate rather than high.

Key data gaps & impacts on confidence:

- **Absence of NiMet hydrological inundation rasters** for the 2027 scenario (prevents precise flood extents & refined exposure estimates).
- **Lack of high-resolution community population rasters** (WorldPop/LGA proxies used — may misallocate population in urban informal settlements).
- **Limited high-resolution DEM/drainage & engineered drainage networks** (SRTM used; does not capture micro-topography or culvert/embankment failures).
- **Incomplete drainage capacity datasets** — drainage vulnerability was scored from qualitative proxies (urban imperviousness, reports of blocked drains); this reduces precision.

6. Assumptions

1. State population = **5,953,500** (2022 projection).
2. Where community population missing, LGA population divided conservatively across communities (documented in CoT).
3. Elevation proxies = SRTM/DEM representative town center elevations. Lower elevation → more vulnerable.

4. Rainfall intensity uses NiMet severity as primary and IMERG/GPM as a spatial proxy for relative intensity (NiMet forecast takes precedence).
5. Drainage capacity scored qualitatively 0–100 based on urbanization/landcover/press reports; 100 = worst drainage.
6. Displacement ratios are binding **10–20%** for high-risk communities; deviations must be justified with new evidence. I did not deviate from the binding range.
7. Minimum camp capacity = **500 persons** as per prompt baseline; hubs assumed 1,000–2,000 for operational mixes.

7. Immediate operational recommendations (what EHOAN / Anambra Government should do now)

1. **Request NiMet hydrological inundation rasters** for all 2027 scenarios (essential for refining inundation extents and exposure).
2. **Rapid GIS validation:** obtain high-resolution population rasters (WorldPop 100m or finer), LiDAR/SRTM/ALOS DEMs if available, and map existing hospitals, schools, roads, and potential camp sites. Update drainage network layers (OSM + local drainage maps).
3. **Pre-identify & sign MOUs** for ~20 hub facilities (hospitals, stadiums, schools) that can be rapidly converted to camps + 100–200 local reception sites on communal land. Prioritize Onitsha & Ogbaru hubs.
4. **Stockpile essentials** in hubs: WASH kits, shelter materials, medical triage kits, and communications gear sized to the 10% baseline first, with contingency plans for 20% surge.
5. **Evacuation routes & logistics:** map and mark primary/secondary evacuation corridors to upland hubs; prioritize route clearance & boat logistics for riverine accesses.
6. **Community messaging & early warning:** strengthen community-level preparedness in the top-7, train ward volunteers, and pre-register vulnerable households.

7. **Refine displacement & exposure models** once NiMet hydrology and higher-resolution DEM/population rasters are available.

Rubrics for Onitsha Flood Forecast

S/N	RUBRICS	M1	M2	M3
1	**[7]** Identify and use a **satellite rainfall dataset** or a **credible proxy source** (e.g., Nigeria Meteorological Agency or NASA Earth Observing System) to determine *rainfall intensity levels* for communities in Anambra State. **Source:** https://gpm.nasa.gov/applications/disasters#:~:text=property)%20they%20impose.-,Satellites%20allow%20us,-to%20monitor%20changes **Quote:** "Satellites allow us to monitor changes in the precipitation, especially over oceans and regions where ground-based data are sparse." **Justification:** IMERG provides half-hourly to daily precipitation estimates and is commonly used to derive relative rainfall intensity and spatial patterns where rain-gauge networks are sparse. Good for relative intensity proxies and trend checks.	0	0	0
2	**[6] Obtain impact & vulnerability records** **Source:** https://dtm.iom.int/sites/g/files/tmzbdl1461/files/reports/Anambra%20Flood%20Assessment%20Dashboard%20%289%20November%202022%29_3.pdf?utm , page 1 **Quote**: "In the seven (7) Local Government Areas (LGAs) of Anambra state assessed, DTM identified 35,074 individuals in 6,980 households that were affected by the floods. These individuals included both IDPs that were living in camp-like settings and residents who were impacted by the floods but remained in their communities." **Justification**: IOM/DTM produced detailed displacement and affected-population dashboards in 2022 for Anambra; NEMA issues national impact summaries. These are the best sources to document historically affected LGAs and validate repeated impact flags for Onitsha & Ogbaru.	0	6	0

3	**[7]** Explicitly extract and apply **population density data at the LGA level** for Anambra State for a **clearly stated reference year**, aligning the dataset's **temporal coverage** with the 2027 flood projection. **Source:** https://www.citypopulation.de/en/nigeria/admin/NGA004__anambra/?utm#:~:text=5%2C953%2C500%20Population%20%5B2022%5D%20%E2%80%93%20Projection **Quote**: 5,953,500 Population (2022) – Projection **Justification:** citypopulation LGA projections were accepted because Worldprop community rasters are missing.	0	7	0
4	**[7]** Explicitly use or reference a **stable, authoritative DEM source** (e.g., NASA SRTM 1 arc-second global DEM) to derive **LGA-level or community-level elevation values**. The output must **state the DEM resolution, vertical datum,** and **processing steps** applied to prepare the elevation proxy used in flood vulnerability ranking. **Source:** https://glovis.usgs.gov/app , https://www.floodmap.net/Elevation/ElevationMap/?gi=2326016#:~:text=Onitsha%2CNigeria%20Elevation%20is%2072%20meter **Quote**: "Onitsha,Nigeria Elevation is 72 meter" **Justification:** SRTM is the standard open global DEM (1″/30m or 1″ non-void)	0	7	0
5	**[4]** Use **both a quantitative proxy** (e.g., impervious-surface / land cover) **and qualitative evidence** (e.g., news or field reports on blocked drainage) to estimate **drainage capacity per community or LGA**. Each component must be explicitly cited and anchored **Source:** https://modis.gsfc.nasa.gov/data/dataprod/mod12.php , https://human-settlement.emergency.copernicus.eu/ **,** https://dailytrust.com/unending-nightmare-of-flooding-in-anambra/?utm#:~:text=The%20only%20drainage%20system **,** **Quote**: "The only drainage system to check flood is the Sacamori drain in Okpoko and it is filled up with sand"	0	4	0

	Justification: Drainage capacity is often not directly available as a dataset. Urbanization / imperviousness, repeated media reports of blocked drains, and local SEMA responses provide evidence for qualitative drainage scoring.			
6	**[8]** Explicitly reference the prompt requirement to identify flood-prone communities and list the **candidate communities (≥10)** it considered with the data basis for each selection (e.g., historical impact records, NiMet mention, NEMA hazard flag, news reports, or repeated DTM/IOM displacement indicators). Separate *selection* from *ranking* and explicitly state the **ranking method** that uses the weighted vulnerability formula from the prompt (Elevation 30%, Population density 25%, Rainfall intensity 25%, Drainage capacity 20%). The ranking step must produce a table of the top **exactly 10** communities sorted by composite vulnerability score **Source:** https://dtm.iom.int/sites/g/files/tmzbdl1461/files/reports/Anambra%20Flood%20Assessment%20Dashboard%20%289%20November%202022%29_3.pdf?utm , page 1. **Quote:** "N/A" **Justification**: IOM/NEMA dashboards and news coverage repeatedly flag these LGAs/communities.	0	0	0
7	**[3]** The model **must** explicitly define and apply a **normalization and scoring method** for all four vulnerability components: Elevation (30%), Population Density (25%), Rainfall Intensity (25%), Drainage Capacity (20%). The scoring process must use **min–max normalization** to scale each component to a common 0–100 range, followed by **weighted aggregation** to produce a composite vulnerability score. **Source**: Prompt Task	0	0	0

	Quote: "based on explicit scoring criteria: Elevation (30% weight), Population density (25% weight), Rainfall intensity (25% weight), Drainage capacity (20% weight)." **Justification**: Weighted composite indices and min–max normalization are standard in hazard ranking literature.			
8	**[5]** Extract and populate all required numeric inputs from authoritative, anchored sources — specifically LGA-level population figures for Anambra State — aligned to the projection year 2025 (or the nearest credible year if 2025 is unavailable, with clear justification). Also document how each numeric value was obtained (table reference, projection method, and year). **Source**: https://www.citypopulation.de/en/nigeria/admin/NGA004__anambra/?utm **Quote**: "N/A" Justification: This is a data-integration step.	0	0	0
9	**[5] Compute a composite vulnerability score** for each community/LGA using the **explicit weighted formula** defined in the prompt: Elevation (30%), Population density (25%), Rainfall intensity (25%), Drainage capacity (20%). The calculation must include: **Intermediate component scores** after normalization**, Weighted aggregation** into the composite score**, Presentation in a structured, reproducible table.** **Source**: Task Prompt **Quote**: N/A **Justification**: The calculation step is deterministic; emphasize digit-by-digit checks and reproducibility.	0	5	0
10	**[9] List exactly 10 communities** and **rank them strictly** in **descending order of composite vulnerability scores** (highest score = rank 1). Present results in a **clearly structured table**	0	0	0

	with the following **minimum columns**: Rank (1–10), Community Name, LGA, Composite Score (numeric, rounded consistently), Component Scores (E, P, R, D). Ties must be handled deterministically (e.g., secondary sort by community name A–Z). **Source**: Prompt Task **Quote**: "Identify and rank the top ten (10) flood-prone communities based on the computed vulnerability scores" **Justification**: Present components so users can audit contributions to composite scores; traceability requires direct citations for each column's data source.			
11	**[6]** I**dentify the LGAs corresponding to the top 10 ranked communities** (from Response #10). **Retrieve population figures** from the Citypopulation LGA table for each of those LGAs, with **explicit citation of the LGA name and year. Sum these population values** to produce the **total exposed population proxy** for the top 10 high-risk communities. Present a **verification table** with at least the following columns: `Community Name`, `LGA`, `Population (Year)`, `Source Anchor (Quote or Table Row)`. Show the **final summed value** clearly beneath the table. **Source:** https://www.citypopulation.de/en/nigeria/admin/NGA004__anambra/?utm **Quote:** N/A **Justification:** The calculation step is deterministic	0	0	0
12	**[6]** The model **must define an explicit, reproducible rule** to label communities as **"high-risk"** based on the computed composite vulnerability scores (from Rubric #9/#7/#10). The rubric accepts one of two permitted, clearly-stated methods — the model must pick one and apply it consistently. Produce a small table showing: Community | Composite Score | High-Risk? (Yes/No) | Method Used (Quartile/Threshold). Use the ranking and computed scores only.	0	0	0

	Source: Prompt task **Quote**: N/A **Justification**: Simple arithmetic using state population projections (5,953,500 projection).			
13	**[9]** Add explicit checks: (a) apply 10–20% ratio to high-risk communities, (b) include camp capacity assumption with minimum 500 per camp, and (c) require justification and stated new ratio if deviating. **Source**: Task Prompt **Quote**: "Estimate the minimum number of Internally Displaced Persons (IDP) camps the Anambra State government should establish in advance of the 2027 flood. Use a binding population displacement ratio of 10–20% for high-risk communities. Justify deviations (if any) with credible evidence and specify the new ratio. Include camp capacity assumptions (e.g., minimum 500 people per camp)." **Justification**: Using a 10–20% displacement range is a defensible operational assumption for flood scenarios if supported by historic DTM displacement percentages.	0	0	0
14	**[9]** Retrieve and cite the exact figure and year of the total state population from Citypopulation.de, anchoring it to the specific table row or section. It must then use the minimum and maximum displaced population figures computed in Rubric #13 and clearly present these values. Using the formula Exposure = $\frac{Displaced\ Population}{Total\ state\ Population} * \frac{100}{1}$ Calculate the minimum and maximum exposure percentages and display them explicitly. It must also provide a short, explicit justification describing the confidence level of the estimate, referencing data quality (such as population reliability and displacement ratio source) and uncertainty factors (such as rainfall variability or drainage conditions). The confidence must be labeled qualitatively as High, Medium, or Low with a brief explanation.	0	0	0

	Finally, the result must be presented in a structured output table with the following columns: Total State Population, Min Displaced, Max Displaced, Min Exposure (%), Max Exposure (%), Confidence Level, and Justification. **Source**: Task Prompt **Quote:** "Quantify expected exposure as a share of the state's population and provide a defensible justification for the rang" **Justification:** Simple arithmetic using state population projections			
15	**[9]** Use the fairness KPI stated in the prompt to ensure that at least 90% of the high-risk population is located within 10 km of at least one proposed IDP camp or designated evacuation center. It must demonstrate this through explicit geospatial or tabular evidence showing the distance coverage and percentage of population reached. Discuss and justify the proximity of each selected site to existing health facilities and assess operational feasibility, including terrain accessibility, transportation infrastructure, and logistical considerations. The output should include a structured table listing each proposed site with its LGA, total high-risk population covered, distance to the site, proximity to the nearest health facility, and operational feasibility notes, accompanied by a brief narrative explanation of how the 90% KPI was achieved. **Source**: Prompt Task **Quote: "**using the following fairness KPI: 1. At least 90% of high-risk population must be within 10 km of at least one IDP camp or designated evacuation center 2. Locations must balance equitable access, coverage, proximity to health facilities, and operational feasibility (terrain, logistics, transport links)." **Justification**: Using 10 km buffer coverage and hub selection is an operational site-siting method.	0	0	0